You have completed a walk-through of a basic blind auction with two bidders and a beneficiary. You can test other cases, such as when a bid is rejected during the `Reveal` phase because it was placed by an imposter who does not know the secret code used when hashing the blind bid. Recall that simple hashing of the parameter value can provide privacy, but hashing the parameters with a secret code provides security. The method used here is not a direct encrypt and decrypt, as you would see in traditional programming. The technique used for hashing in the blind auction illustrates a different approach to privacy and security--one that is appropriate for decentralized blockchain applications.

5.5 Retrospective

In the blind auction problem, the parameters (auction bid details) need to be kept private and secure in one of the phases but can be revealed to the public in another phase. Privacy and security are achieved during the `Bidding` phase by secure-hashing the parameters; when the parameters are revealed during the `Reveal` phase, the bid is verified by computing the hash from the revealed parameters and matching the hash sent during the `Bidding` phase. Thus, the approach used in the blind auction is not implementing security and privacy in the traditional sense of encrypting data with a private key and then decrypting it with a public key.

Other problems have similar hide-and-reveal phases, such as online quizzes and exams, requests for proposals (RFPs) in business contract bidding, and games such as poker (which has play, bet, and reveal phases).

Also, notice that the hashes and identities in the blockchain context use a 256-bit address space and 256-bit computations. This large address space ensures both the uniqueness of decentralized identities and an extremely low probability of collisions for hash values.